

STAT 5640

Capstone in Statistics and Data Science

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Applied Scientist (100.0 / 100)	87.6 / 100

88

CURRENT AVG SCORE
across 6 job roles

99

POTENTIAL SCORE
with recommended additions

+29

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Course provides senior-level and graduate students the opportunity to apply the knowledge, skills, and abilities developed throughout the Statistics and Data Science major. Working in teams, students undertake a data-driven problem presented by domain experts from government, industry, or academia. The course provides valuable real-world experience for students intending to pursue graduate education or technical careers. Topics include team building, problem solving, research methods, project management, data ethics, and clear communication (oral, written, and visual).

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	100.0	100.0	+0.0	LOW
AI Engineer	99.3	100.0	+0.7	LOW
ML Engineer	91.5	100.0	+8.5	LOW
Machine Learning Engineer	89.0	100.0	+11.0	LOW
Data Scientist	82.0	100.0	+18.0	MEDIUM
MLOps Engineer	63.9	92.6	+28.6	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

LLMs & Generative AI

2 of 6 roles

storage

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

MEDIUM

Consider LLMs & Generative AI content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: storage.

GAP KEYWORDS BY ROLE

ML Engineer	91.5 → 100.0 (+8.5 pts)
learning machine models world problems new tools	
Machine Learning Engineer	89.0 → 100.0 (+11.0 pts)
sets learning analytics tools design development models applications	
Data Scientist	82.0 → 100.0 (+18.0 pts)
sets analytics presentation problems	
MLOps Engineer	63.9 → 92.6 (+28.6 pts)
scientists learning machine tools models development	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.